CariSurg MedTech Pathways Programme - Week 0

Student Information

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Missing Clinical Variables in Emergency Triage Dataset: Blood Glucose, Pain Score, Weight, Chief Complaint

The emergency department triage dataset contains essential physiological vital signs such as pulse, temperature, respiratory rate, blood pressure, and oxygen saturation. However, there are several clinically important variables not included in the dataset that would significantly improve patient assessment and triage accuracy.

From my perspective as a medical assistant, one of the most important missing variables is blood glucose level. This is a critical metabolic marker used to identify acute conditions such as hypoglycemia, diabetic ketoacidosis (DKA), and hyperosmolar hyperglycemic state (HHS), which may present with non-specific symptoms but require immediate intervention.

Pain score is also a key missing variable. In clinical practice, pain is routinely assessed using a numeric scale 1 to 10 and is an important indicator of underlying pathology. High pain scores may indicate serious conditions such as appendicitis, renal colic, torsion, or vascular emergencies that may not be immediately evident through vital signs alone.

Patient weight is another important clinical factor, as it directly affects medication dosing, fluid resuscitation calculations, and physiological interpretation of patient status in emergency care settings.

In addition, chief complaints are a critical missing component. Vital signs alone do not provide clinical context, and the reason for presentation is essential for accurate interpretation. For example, tachycardia may be related to anxiety in one case, but may indicate sepsis, hemorrhage, or pulmonary embolism in another.

Including these additional variables would improve clinical decision-making, strengthen triage accuracy, and enhance the performance of AI-driven healthcare models by providing both physiological data and clinical context.